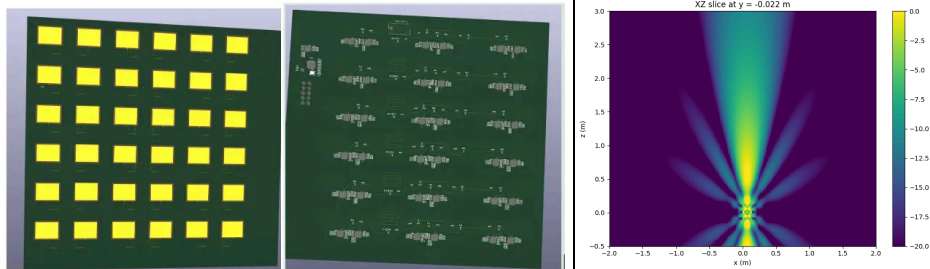


Steerable Microwave Beam Phased Array

Designer: [Levi Sawatzky](#)



1. Project Overview

This project explores the design and construction of a steerable microwave phased array capable of delivering a tightly focused beam of microwave energy toward a target.

The system serves as a science demonstration platform that reveals the power and elegance of microwave circuit design, phased arrays, and high-frequency PCB engineering. The core concept is to electronically steer RF energy without any moving mechanical parts, using precise phase control across an array of patch antennas.

Why This Project Is Interesting / Challenging

- Focused wireless power delivery (e.g., lighting LEDs across a room)
- Electronic beam steering with no moving parts
- Plasma generation inside noble gas bulbs
- Remote microwave heating demonstrations
- Real-world implementation of phased-array beamforming concepts

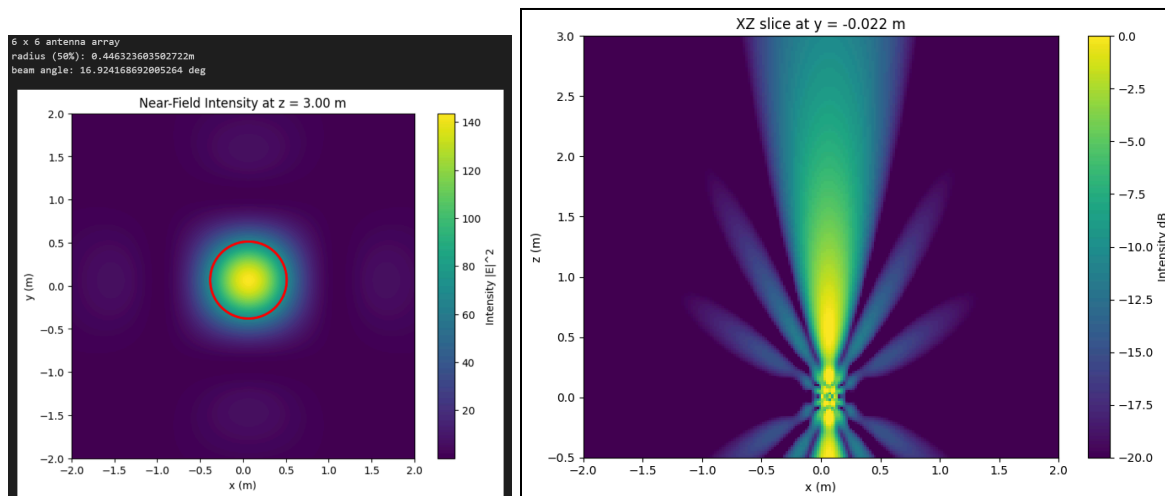
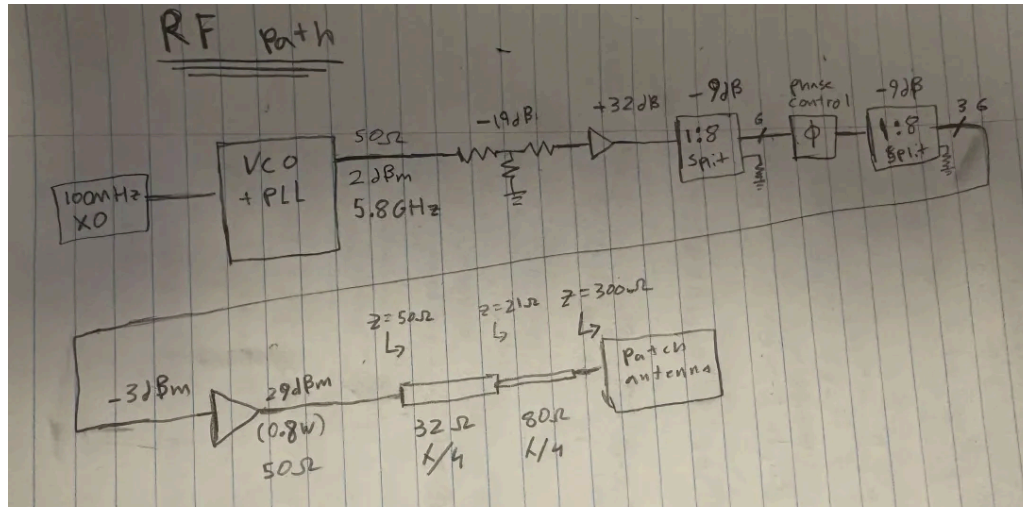
2. System Architecture & Design Approach

The beam is formed using a phased array of microstrip patch antennas. Each antenna element is driven with controlled phase offsets, allowing constructive interference in a desired direction and destructive interference elsewhere.

Key subsystems include:

- RF signal generation and amplification
- Power splitting network
- Phase shifting network

- Patch antenna array layout
- Controlled impedance PCB design



Simulating the beam from my array using python (simplifying the antennas into ideal point sources) reveals a 3dB beam dispersion of 17 degrees.

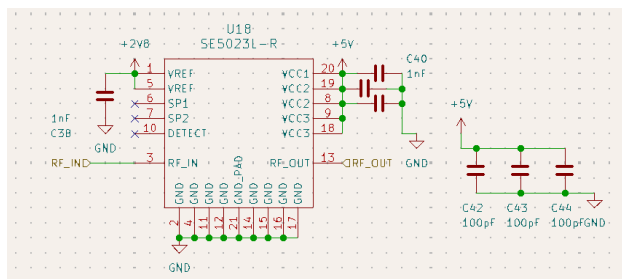
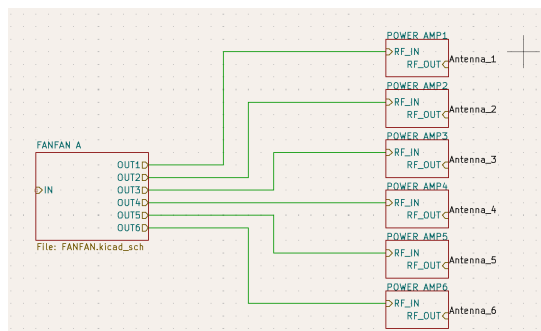
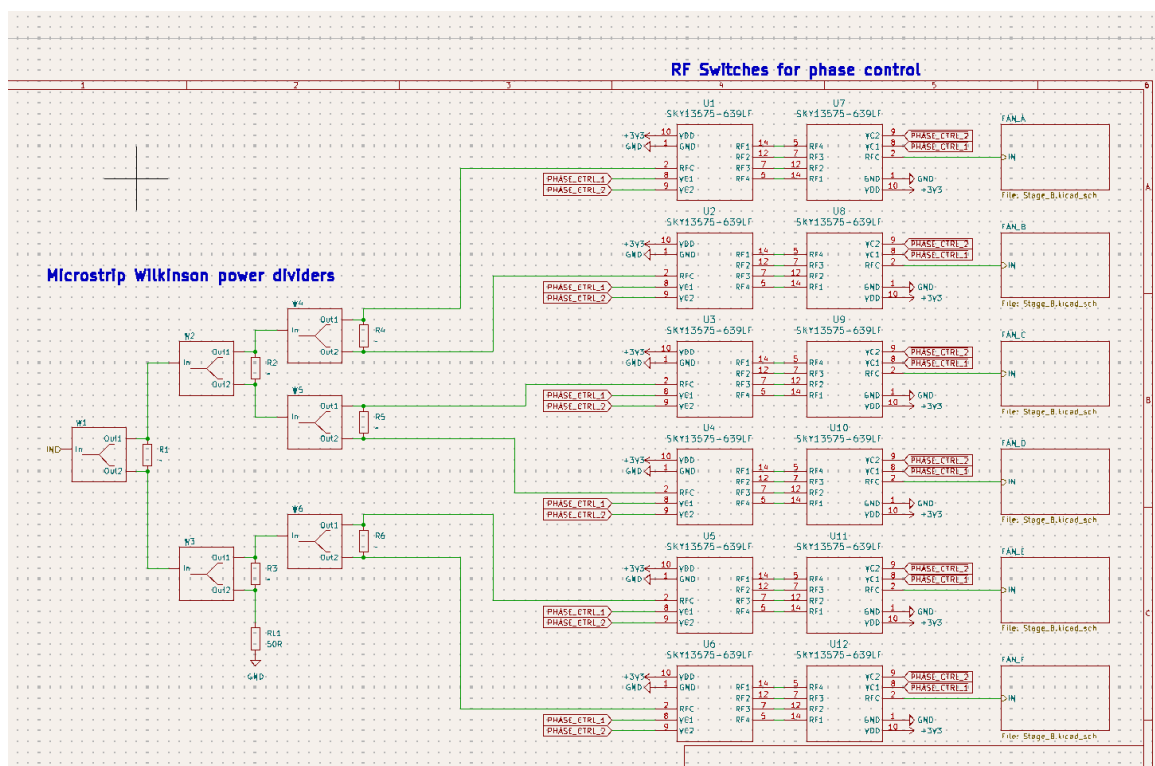
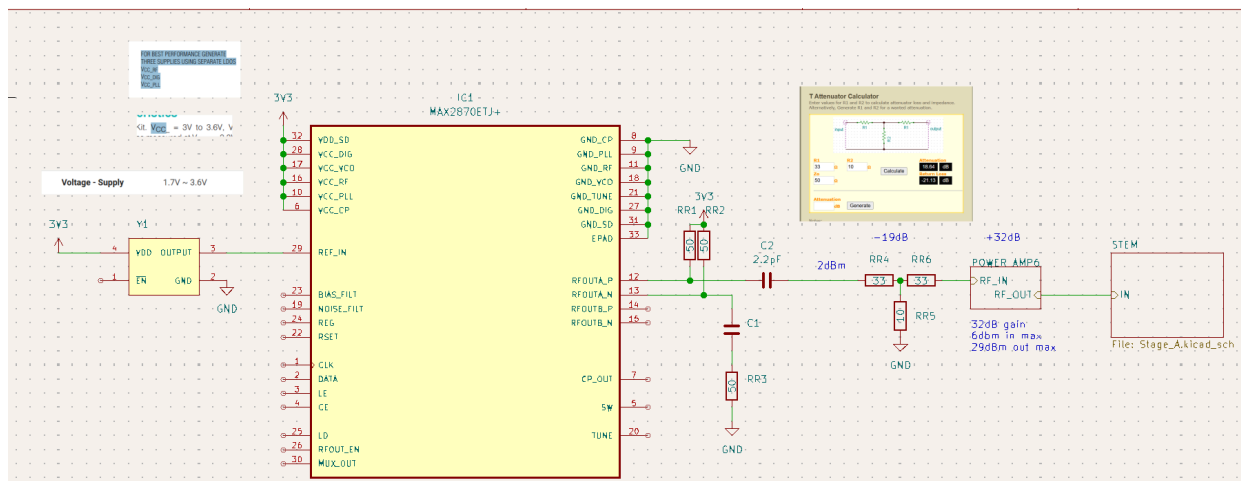
3. Design Process

The design process required careful iteration between simulation and physical layout. Electromagnetic simulations were used to validate antenna resonance, impedance matching, and beamforming characteristics.

Key development steps included:

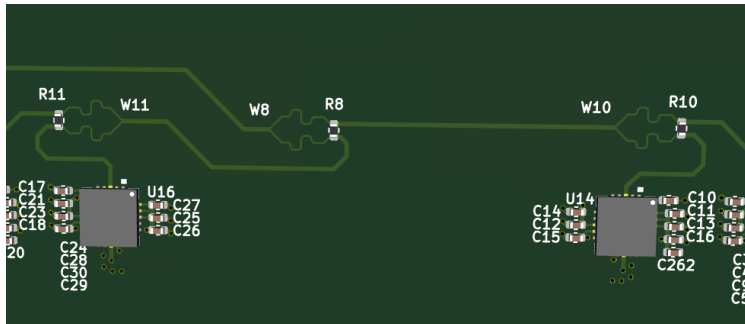
- Patch antenna dimension calculation based on substrate dielectric constant
- Microstrip trace impedance control and stackup selection
- Via modeling and simulation to estimate parasitic inductance effects

- Array spacing optimization to maintain tight focus while allowing a wide range of steerability
- Beam pattern simulation under various phase configurations

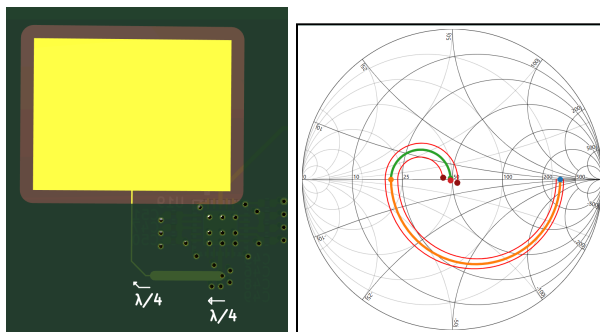


4. Microstrip Impedance Matching

I use **Wilkinson power dividers** to maintain matched impedance in the signal lines as well as provide isolation between the outputs.



Quarter wavelength transformers are used to match the ~300 ohm patch antennas to the 50 ohm amplifiers. QWT was chosen over stub or lumped element matching due to its robustness: 20% inaccuracies in either the length or line impedance still lands with a manageable 10dB return loss.



5. Technical Challenges & Engineering Tradeoffs

- Sourcing RF components that operate reliably at the target microwave frequency
- Managing PCB dielectric tolerances affecting resonance frequency
- Accounting for parasitic inductance and capacitance from vias
- Maintaining consistent phase across RF paths
- Thermal considerations under higher power levels
- Avoiding unintended radiation or coupling between array elements

Most resistors and capacitors do not provide documentation of performance at high frequencies - especially as high as 5.8GHz. Below shows impedance vs frequency charts for resistors chosen for the T attenuator, and capacitors chosen for DC blockers. Murata's "sim surf" website was very handy for choosing appropriate capacitors.

